SIR-Model

Abeli Kai Graham Noirin

Introduction

In this project we consider a spatial version of the Susceptible-Infectious-Recovered (SIR) model. ... Maybe not needed ?

Model Formulation

We consider a spatially distributed population in two dimensions over time.

Variables and Parameters

State variables:

- $S(x, y, t)$: density of susceptible individuals
- $I(x, y, t)$: density of infectious individuals
- $R(x, y, t)$: density of recovered individuals

The local total population densitiy is then given by:

$$N(x, y, t) = S(x, y, t) + I(x, y, t) + R(x, y, t).$$

Parameters:

- $\beta > 0$: transmission rate (time^{-1})
- $\gamma > 0$: recovery rate (time^{-1})
- $D_s \geq 0$: diffusion coefficient of susceptible individuals (space^2/time)
- $D_i \geq 0$: diffusion coefficient of infected individuals (space^2/time)

The PDE System used, represents spatial movement for the infected compartment through the diffusion rate D_i and for the susceptible compartment through D_s.